

HW 6

Exercise 1.

We are estimating the relative error y with a dataset of $n = 49$ observations. 8 independent binary variables were evaluated as potential predictors, which are all formulated as dummy variables. Please answer the following questions. All these questions are abstract questions that no real dataset is provided.

- In Step 1 of the stepwise regression, how many different one-variable models are fit to the data?
- In Step 1, the variable x_1 is selected as the “best” one-variable predictor. How is this determined?
- In Step 2 of the stepwise regression, how many different two-variable models (where x_1 is one of the variables) are fit to the data?
- The only two variables selected for entry into the stepwise regression model were x_1 and x_8 . The stepwise regression yielded the following prediction equation:

$$\hat{y} = .12 - .28x_1 + .27x_8.$$

Give a practical interpretation of the β estimates multiplied by x_1 and x_8 .

- Why should a researcher be wary of using the model, part (d), as the final model for predicting effort (y)?

Exercise 2 (Dataset C.csv).

The dataset C.csv contains 7 independent variables and 1 response variable Y .

- Conduct a stepwise regression analysis of the data using an available statistical software package.
- Interpret the β estimates in the resulting stepwise model.
- What are the dangers associated with drawing inferences from the stepwise model?

Exercise 3 (Dataset G.csv).

Consider the dataset G.csv. The response variable is y .

- Use stepwise regression (with stepwise selection) to find the best predictors of y .
- Use stepwise regression (with backward elimination) to find the best predictors of y .
- Use all-possible-regressions-selection to find the best predictors of y .

- d. Compare the results, parts a-c. Which independent variables consistently are selected as the best predictors?
- e. Explain how you would use the results, parts a-c, to develop a model for y .